

Advanced (Habanero)

Q1. What is the standard form of a quadratic function?

- (A) $ax^2 + bx + c$
 (B) $ax + bx^2 + c$
 (C) $ax^2 - bx + c$
 (D) $ax^3 + bx^2 + c$
 (E) $ax + b$

$x^2 + 2x + 1?$

- (A) $x = -1$
 (B) $x = 1$
 (C) $x = 0$
 (D) $x = -2$
 (E) $x = 2$

Q2. Which of the following is a quadratic function in standard form?

- (A) $x^2 + 3x - 2$
 (B) $x^3 - 2x^2 + x$
 (C) $2x^2 - 5x + 1$
 (D) $x^4 - 2x^3 + x^2$
 (E) $x - 2$

Q6. Which of the following is a key feature of the quadratic function $x^2 + 2x + 1$?

- (A) Vertex at $(0, 0)$
 (B) Axis of symmetry at $x = -1$
 (C) Minimum value at $x = -1$
 (D) Maximum value at $x = 1$
 (E) All of the above

Q3. What is the value of a in the quadratic function $2x^2 + 3x - 1$?

- (A) 1
 (B) 2
 (C) 3
 (D) -1
 (E) -2

Q7. What is the minimum value of the quadratic function $x^2 + 2x + 1$?

- (A) 0
 (B) 1
 (C) -1
 (D) -2
 (E) 2

Q4. Which of the following quadratic functions has a vertex at $(0, 0)$?

- (A) $x^2 + 2x + 1$
 (B) $x^2 - 2x + 1$
 (C) $x^2 + 1$
 (D) $x^2 - 1$
 (E) x^2

Q8. Which of the following quadratic functions has a maximum value?

- (A) $x^2 + 2x + 1$
 (B) $-x^2 + 2x + 1$
 (C) $x^2 - 2x + 1$
 (D) $x^2 + 1$
 (E) $x^2 - 1$

Q5. What is the axis of symmetry of the quadratic function**Open Ended Questions (FRQ)****Q9.** Find the vertex and axis of symmetry of the quadratic function $x^2 - 4x + 3$. Show all work and explain your reasoning.**Q10.** Graph the quadratic function $x^2 + 2x - 1$ and identify its key features. Use the graph to find the minimum value of the function.

Chef's Tips

- Provide clear instructions and examples before the test.
- Emphasize the importance of showing work and explaining reasoning.
- Allow students to use graphing calculators or software to visualize the quadratic functions.